

SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID –**
- **Problem Statement Title-**
- **Theme-**
- **PS Category- Software/Hardware**
- **Team ID-**
- **Team Name (Registered on portal)**



THE PROBLEM- Traditional vital monitoring requires physical contact like cuffs or sensors, causing healthcare bottlenecks. This equipment is often unavailable, slows outpatient triage, and is unfeasible for remote telemedicine.

OUR IDEA- HemoVision uses standard webcams or phones to extract vital signs contactlessly from video. Deep learning reconstructs pulses from skin color alone, enabling zero-cost, scalable mass screenings.

TRADITIONAL MONITORING

- Requires physical contact - cuffs, clips, wired sensors
- Equipment isn't always available at point of care
- Slows down outpatient triage at high volumes
- Uncomfortable for continuous / long-term monitoring
- Not feasible for remote / telemedicine screening

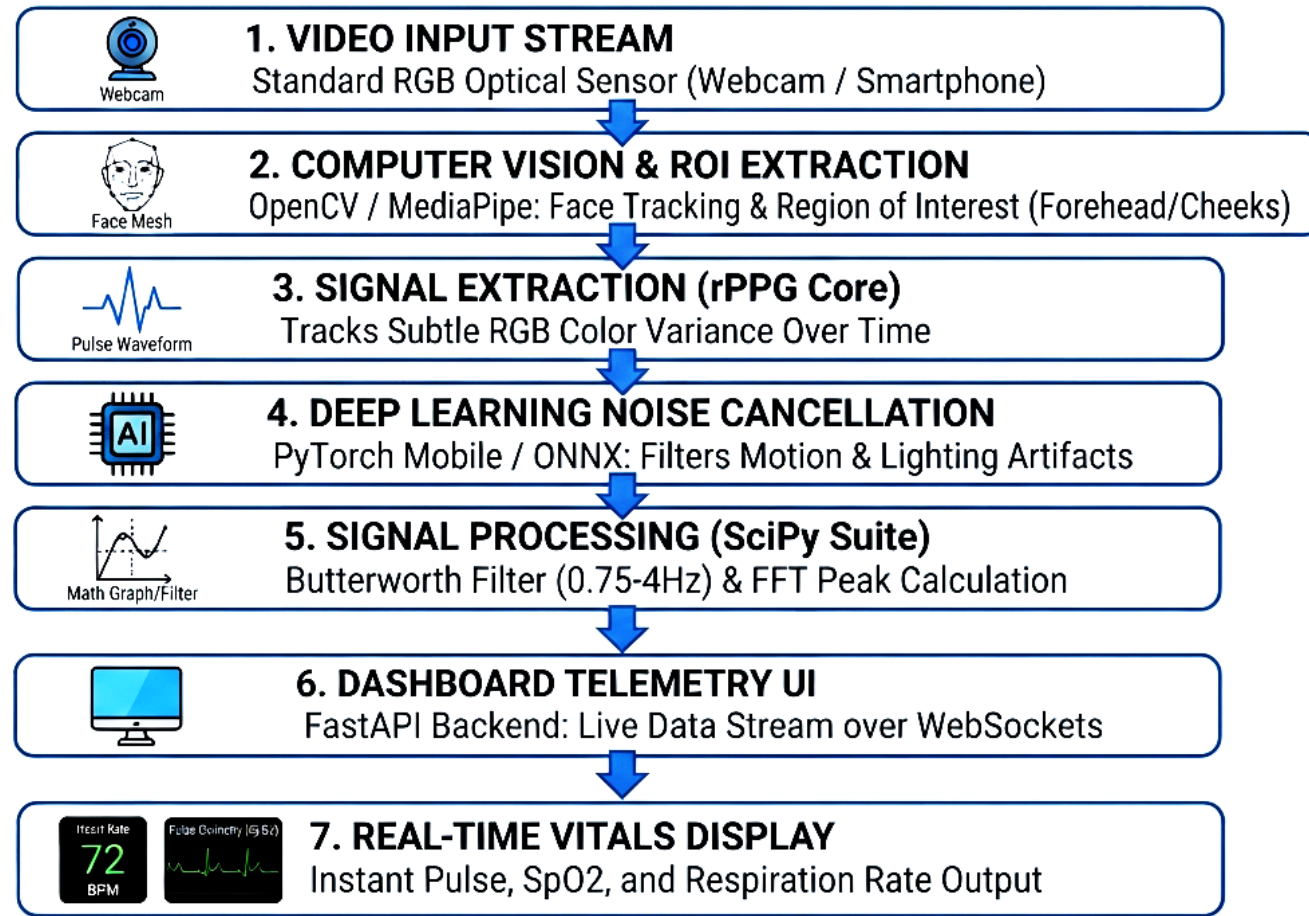
OUR APPROACH: HEMOVISION

- Just a face on ordinary video - no sensor to attach
- Works with any standard camera - webcam or phone
- Deep learning reconstructs the pulse from color alone
- Non-invasive, instantaneous, zero additional cost
- Scalable to mass screening & telemedicine

Hardware & Software-

- **1080p Webcam (Logitech C920):** Captures high-quality facial video to track subtle skin color changes.
- **NVIDIA Jetson Nano / Raspberry Pi 5:** Runs the AI models and processes data locally at the edge.
- **Li-ion Power Bank (20,000 mAh):** Provides portable, continuous battery power for remote medical setups.
- **5-inch Touchscreen LCD:** Displays real-time heart rate graphs and patient vital signs on screen.
- **Python / NumPy / SciPy:** Handles data math, filters out noise, and calculates final heart rate values.
- **OpenCV / Google MediaPipe:** Tracks 468 facial points instantly to lock onto the forehead and cheeks.
- **PyTorch Mobile / ONNX Runtime:** Runs lightweight AI models to remove movement and lighting errors.
- **FastAPI / WebSockets:** Streams video data and patient results to the display dashboard instantly.

HEMOVISION: SYSTEM ARCHITECTURE & TECHNICAL FLOW



Feasibility-

- **Software-driven deployment:** Runs entirely on standard RGB cameras, eliminating initial specialized medical hardware dependencies.
- **Edge-compute compatibility:** Operates locally on low-cost units like Raspberry Pi 5 without requiring expensive cloud servers.
- **Simulated testing environment:** Uses pre-recorded public facial video datasets to validate accuracy safely before real-time testing.
- **Open-source foundation:** Built on stable frameworks like OpenCV and PyTorch, ensuring high flexibility and zero licensing costs.

Challenges-

- **Ambient illumination noise:** Changing indoor lighting or flickering fluorescent bulbs can corrupt subtle skin color signals.
- **Subject motion artifacts:** Sudden head movements or talking causes the face tracker to lose the specific skin regions.
- **Clinical validation barriers:** Proving consistent, medical-grade accuracy across diverse skin tones to secure regulatory approvals.

Commercial feasibility-

- **Zero hardware overhead:** Enables hospitals to upgrade to contactless vitals tracking using their existing webcams.
- **B2B SaaS subscription model:** Targets steady, recurring revenue from telemedicine platforms, smart clinics, and remote health apps.
- **High-margin scalability:** Software-only distribution ensures near-zero replication costs and rapid global market expansion.
- **Reduced hospital operational costs:** Drastically shortens outpatient triage times, allowing clinics to handle higher patient volumes with fewer physical sensors.

Strategy-

- **Deep learning filtration:** Trains the PyTorch core model to actively filter out sudden shadows and ambient light flashes.
- **Dynamic ROI tracking:** Uses MediaPipe's 468-point mesh to lock onto the forehead and cheeks, even during movement.
- **Phased clinical partnerships:** Teams up with local clinics to run parallel validation tests alongside standard pulse oximeters.

Direct Impact on Target Users

- **Instant Access:** Users check their heart rate immediately using their current phone or computer webcam without buying extra gear.
- **Effortless Monitoring:** Tracks vitals automatically and quietly in the background during normal screen time with zero setup.
- **Zero Discomfort:** Provides a completely touchless experience that eliminates skin irritation from tight straps or sticky patches.
- **No Battery Hassles:** Eliminates the daily chore of charging, updating, or syncing another physical wearable device.
- **Early Peace of Mind:** Detects subtle health changes early from home, helping users know exactly when to consult a doctor.

Strategic Impact

- **Market Disruption:** Challenges expensive wearables by turning standard webcams into health monitors.
- **Frictionless Scaling:** Bypasses hardware supply chains to deploy instantly worldwide via software.
- **Telehealth Expansion:** Embeds real-time vital tracking directly into standard video clinic calls.

Economic & Strategic Benefits-

- **Massive Cost Savings:** Eradicates the need to buy, maintain, or replace expensive hardware wearables and medical sensors.
- **Instant Global Reach:** Deploys immediately to millions of users via the web, completely bypassing manufacturing and shipping bottlenecks.
- **Low Operational Overhead:** Operates as a pure software-as-a-service (SaaS) model, drastically lowering production and distribution costs.
- **High Market Disruption:** Weakens the market hold of traditional fitness trackers by transforming everyday webcams into health tools.
- **B2B Revenue Streams:** Opens lucrative licensing opportunities with virtual clinics, workplace wellness programs, and insurance apps.

Your
Team
Name

RESEARCH AND REFERENCES



- Details / Links of the reference and research work